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A two-stage model for stock price prediction based on variational mode decomposition and ensemble machine learning method

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decomposition (VMD). Subsequently, three individual machine learning models, namely support vector machine regression (SVR), extreme learning machine (ELM), and deep neural network (DNN), are separately employed to predict decomposed sub-series, and then the obtained sub-series predictions of each individual prediction model are aggregated to generate the preliminary stock price predictions. In the second stage, an ELM-based nonlinear ensemble strategy is employed to combine preliminary stock price predictions. To verify the effectiveness of the proposed two-stage model, it is compared with a total of fourteen models in terms of accuracy evaluation, improvement percentage comparison, and statistical test. The empirical results demonstrate that the proposed two-stage model can obtain better performance than other competitor models.

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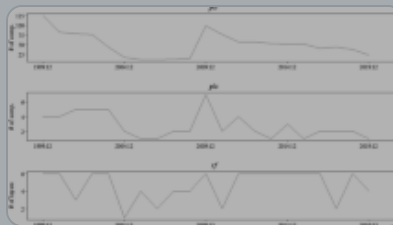
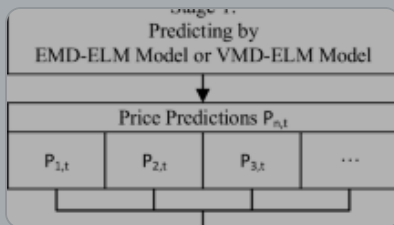
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Data Availability

The data of this study are from Wind database.

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Contributions

JZ contributed to conceptualization, writing—review and editing, and supervision. XC contributed to conceptualization, methodology, software, and writing—original draft.

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